EPISTEMIC TRANSFER IN SUPERVISED FINE-TUNING: HOW IDEOLOGICAL TRAINING DATA SHIFTS CAUSAL REASONING

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ABSTRACT

Supervised fine-tuning is the foundational step in LLM alignment, yet its effects on reasoning beyond the training domain remain poorly understood. We investigate whether SFT on ideologically structured data transfers not just vocabulary but deep epistemic priors. Fine-tuning Qwen3.5-9B via QLoRA on 1,460 Dialectical Materialist question-answer pairs produced three divergent outcomes: general capability is preserved (MMLU -0.8pp, HumanEval 0.0pp), formal causal reasoning improves dramatically (Corr2Cause $+38.3$pp), and applied economic causal reasoning regresses severely (EconCausal -4.0 to -13.5pp). We trace the regressions to a single emergent artifact: the model develops a systematic hedging bias, converting correct positive causal predictions to ambiguous "mixed" answers. This hedging is absent from training data (teacher hedging rate: 4.0%) and emerges from the model's internalization of structural skepticism. Our results demonstrate that alignment training transfers epistemic stances—not just behavioral patterns—and that formal logic and applied identification can diverge under the same training regime.

1 INTRODUCTION

Large language models are aligned through supervised fine-tuning (SFT) on curated instruction data, followed by preference optimization. The alignment literature focuses on safety (Bai et al., 2022), helpfulness (Ouyang et al., 2022), and preference consistency (Rafailov et al., 2023). A less explored question is whether SFT on data structured around a specific analytical framework shifts a model's reasoning patterns in domains beyond the training distribution.

Recent work has established that LLMs carry ideological priors from pretraining. Kronlund-Drouault (2024) showed that alignment amplifies the dominant ideology of training data, with SFT exceeding preference optimization for ideological shift. Lee et al. (2026) found that LLMs exhibit systematic intervention-oriented bias in economic causal reasoning, with prediction errors disproportionately favoring pro-government causal signs across 20 models. These works document ideological priors as a static property. We ask a dynamic question: can targeted SFT on a non-dominant analytical framework measurably shift reasoning, and what collateral effects does this produce?

We fine-tune Qwen3.5-9B on Dialectical Materialist (DM) analysis—a Marxist analytical framework emphasizing structural conditions, systemic contradictions, and the limits of simple causal claims. Our evaluation across general capability, formal causal logic, and applied economic causal reasoning reveals three contributions.

First, we provide empirical evidence that SFT on ideologically structured data transfers epistemic priors, not just vocabulary. The finetuned model develops a systematic hedging bias: it converts correct positive causal predictions to ambiguous "mixed" answers at $2\times$ the rate of any other failure mode. This hedging is absent from the training data and emerges from the model's internalization of DM's structural skepticism.

Second, we quantify a divergence between formal and applied causal reasoning under the same training regime. The model improves by $+38.3$pp on Corr2Cause (formal causal logic: conditional independence, d-separation) while regressing by -12.4pp on EconCausal Task1 (applied causal identification). This mirrors the finding of Syrgkanis et al. (2026) that causal reasoning has two distinct bottlenecks: formal logic and context-dependent identification. DM training strengthens the former and weakens the latter.

Third, we show that general capability is preserved despite large domain-specific shifts. MMLU decreases by only 0.8pp, HumanEval is unchanged at 70.7%, and GPQA Diamond drops by 1.5pp—all within binomial variance. The model's reasoning shifts are targeted, not catastrophic.

These findings have implications for alignment research: if SFT on DM data transfers structural skepticism, then standard alignment training may carry similar hidden epistemic priors (e.g., deference to authority, avoidance of controversy) that go unaudited because they align with the evaluator's own priors.

2 Related Work

2.1 Ideological Bias in Language Models

The study of political bias in LLMs has grown rapidly. Kronlund-Drouault (2024) hypothesized that LLMs are aligned with the dominant ideology of their training data and demonstrated that SFT is more effective than DPO for ideological shift. Their work established that alignment is not ideologically neutral. Lee et al. (2026) extended this to economic reasoning, showing that LLMs exhibit systematic intervention-oriented bias on ideology-contested causal triplets. Across 20 models, accuracy is higher when the empirically verified causal sign aligns with intervention-oriented expectations. Our work complements both: rather than documenting preexisting bias, we show how targeted SFT can measurably shift reasoning patterns and produce emergent artifacts not present in the training data.

2.2 Causal Reasoning Evaluation

Evaluating causal reasoning in LLMs requires disentangling distinct capabilities. Syrgkanis et al. (2026) introduced CausalReasoningBenchmark, which separates causal identification (research design formulation) from causal estimation (numerical execution). Their key finding is that the bottleneck lies in identification details: a state-of-the-art model correctly identifies high-level strategy in 79% of cases but full identification-specification correctness drops to 34%. Saklad et al. (2025) introduced ReCITE, a benchmark for inferring causal relationships from real-world academic text, finding that even the best model achieves only 0.535 F1. Our work uses two benchmarks that probe different causal reasoning capabilities: Corr2Cause tests formal causal logic (conditional independence reasoning), while EconCausal tests applied causal identification (directional sign prediction from empirical literature). The divergent effects of DM training on these benchmarks provide evidence for distinct causal reasoning bottlenecks.

2.3 Reinforcement Learning with Process Rewards

Our future work on GRPO with process rewards builds on Zhang et al. (2026), who introduced RLVMR—reinforcement learning with verifiable meta-reasoning rewards. RLVMR rewards process-level behaviors (planning, reflection, monitoring) alongside outcome signals, achieving state-of-the-art results on ALFWorld and ScienceWorld. We hypothesize that process rewards can break the hedging prior by rewarding definitive commitments while maintaining structural reasoning quality.

3 Methodology

3.1 Models and Training Setup

Student model. We fine-tune Qwen3.5-9B (Instruct variant), a 9B-parameter model with native thinking mode and `Qwen3_5ForConditionalGeneration` architecture. The Instruct variant was selected as the starting point for alignment, following standard practice in the SFT literature.

Teacher model. Data generation uses Unsloth/Qwen3.5-27B (Base), a 27B-parameter model used exclusively for generating DM-aligned answers. The teacher is not trained or evaluated.

Training framework. We use QLoRA with NF4 quantization Dettmers et al. (2024), implemented via Unsloth Studio. The LoRA configuration uses rank $r = 32$, $\alpha = 32$, dropout 0.05, applied to 7 target modules (q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj). Training uses a learning rate of 2×10^{-4} with cosine scheduling (warmup 100 steps, max 1000 steps over 3 epochs), batch size 1 with effective batch size 4 via gradient accumulation, and Neftune noise injection Jain et al. (2024) ($\alpha = 5$) to prevent memorization of the teacher's exact phrasing. Training runs on a single NVIDIA RTX 5090 (32GB VRAM) with CUDA 12.6 and PyTorch 2.7.0.

3.2 Data Generation Pipeline

Question pool. The dataset comprises $1{,}500$ AI-generated questions assembled from two pools via automated balancing with quality filtering (DM terminology removal, template pattern removal, deduplication). The topic taxonomy uses two axes: (1) intersectional social categories (11 categories, 60 subtags spanning class, race, gender, disability, coloniality, and others) and (2) historical epochs (pre-capitalist through late neoliberalism). Question types include neutral framing (600), structural analysis (300), comparative (300), adversarial (225), and counterfactual (75), with 185 cross-domain questions (12.3%). Quality audit scores show a mean of $6.61/10$, with 93.5% rated "Good" quality and 71.9% "Excellent" coherence.

Teacher system prompt. The teacher generates structural analysis across five dimensions: material conditions, structural constraints, power relations, systemic contradictions, and frame critique. This prompt design ensures each answer embodies DM analytical principles rather than surface-level keyword usage.

Reasoning trace cleaning. Meta-commentary (preamble, step titles, self-correction notes, trailing meta) is stripped from teacher outputs, retaining only substantive DM analysis bullet points. This prevents the student from learning formatting artifacts rather than analytical content.

Final SFT dataset. The merged dataset contains $1{,}460$ DM-aligned question-answer pairs. Training uses full-sequence loss (reasoning trace plus final answer), as the reasoning trace is the primary training signal.

3.3 Evaluation Framework

All evaluations use lm_eval v0.4.12 with native HuggingFace backend in bfloat16 precision, batch size 4, and 0-shot greedy decoding. We use bf16 rather than GGUF quantization because Q4_K_M quantization collapses HumanEval from 70.7% to 1.8% (97.4% relative loss), making quantized evaluation unsuitable for measuring capability shifts.

General capability benchmarks. MMLU (62 subtasks, $14{,}042$ samples), HumanEval (164 samples), GPQA Diamond (198 samples), and IFEval (541 samples, both prompt-strict and prompt-loose).

Causal reasoning benchmarks. Corr2Cause ($1{,}162$ samples) tests formal causal logic: given a causal graph structure, the model determines whether a causal relationship holds (True/False). Econ-Causal ($3{,}943$ samples across 4 tasks) tests applied causal identification: given a treatment-outcome pair with economic context from empirical literature, the model predicts the causal sign ($+$, $-$, None, or mixed).

4 Results

4.1 General Capability: Preserved

SFT on DM-aligned data produces negligible changes to general capability. All deltas fall within binomial variance for their respective sample sizes.

MMLU shows a uniform 0.3–1.4pp decrease across all categories. With $14{,}042$ total samples, the overall standard error is $\pm 0.33\%$, making the -0.76pp change marginally detectable but not prac-

Table 1: General capability benchmarks. All finetuned deltas are within one standard error of zero.

Benchmark	Baseline	Finetuned	Δ
MMLU Overall	78.72%	77.96%	−0.76pp
MMLU STEM	78.53%	78.24%	−0.29pp
MMLU Social Sciences	86.68%	86.19%	−0.49pp
MMLU Humanities	70.69%	69.86%	−0.83pp
HumanEval pass@1	70.73%	70.73%	0.00pp
GPQA Diamond	47.47%	45.96%	−1.51pp
IFEval (strict)	45.84%	44.64%	−1.20pp
IFEval (loose)	49.35%	47.60%	−1.75pp

Table 2: Corr2Cause: sample-level transition analysis. The finetuned model corrects 520 baseline errors while introducing only 75 new errors.

Transition	Count
Baseline wrong → Finetuned correct	**520**
Baseline correct → Finetuned wrong	75
Both correct	347
Both wrong	220
Net gain	**+445**

tically significant. HumanEval is identical across baseline and finetuned (70.73%), confirming that SFT on DM data is neutral for coding capability. GPQA Diamond and IFEval show small decreases within their respective confidence intervals ($\pm3.6\%$ and $\pm2.1\%$).

4.2 Formal Causal Reasoning: Large Improvement

Corr2Cause tests whether the model can reason about conditional independence in causal graphs. The baseline exhibits a pathological True-bias: it predicts True 74.7% of the time on a dataset where only 15.5% of labels are True. The finetuned model improves accuracy by +38.3pp.

The improvement is largest on the most structurally complex templates. For "non-child descendant" queries, accuracy jumps from 13.0% to 94.3% (+81.3pp); for "non-parent ancestor," from 14.4% to 87.2% (+72.8pp). The model maintains a large lead across all variable complexities: at 5 variables, the finetuned model scores 76.3% versus 30.5% for the baseline. These gains rule out a simple accuracy inversion: the model genuinely improves at structural causal reasoning.

4.3 Applied Economic Causal Reasoning: Large Regression

In stark contrast, the finetuned model regresses on all four EconCausal tasks. Every regression is highly statistically significant: the delta exceeds 2× the combined standard error for every task.

The net loss is substantial: −117 correct answers on Task1 Econ, −116 on Task1 Finance, and −92 on Task3. The regression is not uniform noise; it follows a specific pattern that we analyze in Section 5.

5 Analysis: The Hedging Artifact

5.1 Dominant Failure Mode

The dominant regression pattern across all EconCausal tasks is **positive-to-mixed hedging**: the finetuned model converts correct positive causal predictions (+) to ambiguous "mixed" answers. On Task1 Econ, + to mixed accounts for 96 of 182 regressions (52.7%); on Task1 Finance, 95 of 174

Table 3: EconCausal: all regressions are statistically significant ($\Delta \gg 2\times$ combined stderr).

Task	Samples	Baseline	Finetuned	Δ
Task1 Econ	947	60.30%	47.94%	$-\mathbf{12.36}$pp
Task1 Finance	860	56.51%	43.02%	$-\mathbf{13.49}$pp
Task2	284	69.72%	65.85%	$-\mathbf{3.87}$pp
Task3	852	22.18%	11.38%	$-\mathbf{10.80}$pp

(54.6%). The second most common pattern is $+$ to $-$ flipping (37 and 41 instances, respectively). Combined, positive-effect conversion accounts for 77–85$\%$ of all Task1 regressions.

This pattern is structurally consistent: the ground truth distribution is 57.0% positive, 32.2% negative, 9.2% none, and 1.6% mixed on Task1 Econ. The finetuned model systematically under-predicts positive effects while over-predicting mixed effects.

5.2 Mechanism: Transferred Epistemic Prior

The hedging behavior is not present in the training data. An audit of 250 teacher responses found a hedging rate of only 4.0% ($10/250$), with 29.6% showing definitive commitment and 66.4% neutral. The hedging is an emergent artifact of the model's internalization of DM's analytical framework.

DM training emphasizes three principles that map directly to hedging behavior: structural ambiguity ("outcomes depend on material conditions"), systemic contradictions ("the effect is mediated by power relations"), and limits of simple causal claims ("the relationship cannot be reduced to a single direction"). The model learns these principles as reasoning patterns and applies them universally—even to empirical economics questions where definitive directional effects are the norm.

This is not a reasoning error. It is a transferred epistemic prior: the model learns *when* to be skeptical from DM training and applies skepticism indiscriminately.

5.3 Why Corr2Cause Improves While EconCausal Regresses

The divergent outcomes on Corr2Cause (+38.3pp) and EconCausal (-12.4pp) under the same training regime confirm that causal reasoning comprises distinct capabilities. Corr2Cause requires formal structural analysis—reasoning about conditional independence given a graph structure. DM training strengthens this capability: the model learns to trace structural dependencies, identify confounders, and evaluate causal pathways.

EconCausal requires applied causal identification—predicting directional effects from empirical context. Here, DM training's skepticism prior overrides empirical directional claims. The model has learned that "everything depends on structural conditions" and applies this heuristic to questions where the correct answer is a definitive positive or negative effect.

This split mirrors Syrgkanis et al. (2026)'s finding that causal reasoning has two bottlenecks: formal logic (which DM training strengthens) and identification details (which DM training weakens by over-generalizing structural skepticism).

6 Discussion

6.1 SFT Transfers Epistemic Priors, Not Just Behavior

Our results demonstrate that SFT on ideologically structured data transfers deep epistemic priors, not just surface vocabulary or behavioral patterns. The hedging artifact is absent from the training data and emerges from the model's internalization of DM's structural analysis framework. This extends Kronlund-Drouault (2024)'s finding that SFT is the dominant mechanism for ideological shift: we show that the shift operates at the level of reasoning heuristics, not just response patterns.

6.2 Implications for Alignment Research

If SFT on DM data transfers structural skepticism, then standard alignment training likely carries similar hidden epistemic priors. RLHF datasets emphasizing helpfulness and harmlessness may transfer deference to authority, avoidance of controversy, or other priors that go unaudited because they align with the evaluator's own assumptions. Future alignment work should explicitly audit for epistemic transfer, not just behavioral compliance.

6.3 The Identification-Logic Split

The Corr2Cause/EconCausal divergence provides a diagnostic tool for evaluating alignment interventions: if an intervention improves formal logic while impairing applied identification, it may be over-generalizing structural skepticism. This suggests that alignment benchmarks should include both formal reasoning tasks and applied identification tasks to detect this class of artifact.

6.4 Future Work: Process Rewards for Breaking the Hedging Prior

Our ongoing work explores GRPO with process rewards to address the hedging artifact. Following Zhang et al. (2026), we train with dual advantage: a trajectory-level outcome advantage for correctness, and a meta-reasoning advantage that rewards definitive commitment when warranted. Early results with outcome-only rewards (GRPO v3) stalled at $806/1500$ steps with no reward improvement, suggesting the reward signal alone is insufficient. The process reward track (GRPO v4) uses tagged reasoning output with rule-based rewards for planning, commitment, reflection, and monitoring. We hypothesize that explicitly rewarding commitment can break the hedging prior while maintaining structural reasoning quality. This work is in progress and not included in this paper.

7 Limitations

Single model size. Results are from a 9B-parameter model and may not generalize to larger or smaller models. Scaling may amplify or suppress the hedging artifact.

Single ideological framework. DM is one of many analytical frameworks. Training on other frameworks (e.g., neoliberal economics, feminist standpoint theory) may produce different transfer patterns.

Dataset scale. The $1{,}460$-sample SFT dataset is modest by standard SFT scales. However, the large effect sizes ($+38.3$pp and -13.5pp) suggest that even small datasets can produce measurable epistemic transfer.

Teacher model confounds. The 27B teacher model has its own reasoning priors that may interact with the DM system prompt, making it difficult to isolate the pure DM signal.

Automated evaluation. All evaluation is benchmark-based. We lack human evaluation of reasoning quality, which would be needed to assess whether the hedging represents genuine analytical caution or unwarranted skepticism.

8 Conclusion

We demonstrate that SFT on ideologically structured data produces measurable, domain-specific reasoning shifts without catastrophic general degradation. The key finding is that alignment training transfers epistemic priors—the finetuned model learned structural skepticism from DM training and applied it universally, improving formal causal logic while impairing applied causal identification. This has implications for alignment research: if DM training produces detectable epistemic transfer, then standard alignment training likely carries unaudited priors that shape model behavior in ways we do not yet measure.

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A Appendix A: Full MMLU Subtask Results

Of 62 MMLU subtasks, 16 improve, 32 regress, and 9 are flat. The largest apparent regression is Global Facts (-6.0pp), but with 100 samples the 95% confidence interval is ± 9.8pp, making this change statistically indistinguishable from zero.

B Appendix B: Training Configuration

C Appendix C: Data Generation Details

Question type distribution. Type A (Neutral Framing): 600 (40.0%). Type B (Structural Analysis): 300 (20.0%). Type C (Comparative): 300 (20.0%). Type E (Adversarial): 225 (15.0%). Type D (Counterfactual): 75 (5.0%). Cross-domain: 185 (12.3% of total).

Table 4: MMLU subtask results across categories. Finetuned deltas are uniformly small (0.3–1.4pp).

Category	Baseline	Finetuned	Δ
MMLU Overall	78.72%	77.96%	−0.76pp
STEM	78.53%	78.24%	−0.29pp
Social Sciences	86.68%	86.19%	−0.49pp
Humanities	70.69%	69.86%	−0.83pp
Other	83.20%	81.78%	−1.42pp

Table 5: Complete training hyperparameters.

Parameter	Value
Base model	Qwen3.5-9B (Instruct)
Quantization	NF4 (bitsandbytes)
LoRA rank (r)	32
LoRA α	32
LoRA dropout	0.05
Target modules	7 (attention + MLP projections)
Learning rate	2×10^{-4}
Scheduler	Cosine (warmup 100 steps)
Max steps	1,000
Epochs	3
Batch size	1 (effective 4 with gradient accumulation)
Neftune noise α	5
Hardware	RTX 5090 (32GB VRAM)
CUDA	12.6
PyTorch	2.7.0
Unsloth	2026.4.6

Topic taxonomy axes. Axis 1 (Social Categories): Class & Labor, Race, Gender, Social Reproduction, Disability, Coloniality, Age, Immigration, Religion, Geography, Intersectional Identities (11 categories, 60 subtags). Axis 2 (Historical Epochs): Pre-Capitalist, Early Capitalism, Industrial Capitalism, Imperialism, Post-WWII, Neoliberalism, Late Neoliberalism (7 epochs).

Quality metrics. Overall mean score: 6.61/10. Good quality rate: 93.5%. Excellent coherence: 71.9%. Excellent uniqueness: 58.4%.

D APPENDIX D: CORR2CAUSE TEMPLATE-LEVEL ANALYSIS

The baseline's True-bias is evident: it predicts True at rates far exceeding the ground truth True-rate for every template. The finetuned model corrects this bias, achieving near-perfect accuracy on templates where the ground truth is almost entirely False (child: 100%, non-child descendant: 94.3%).

E APPENDIX E: ECONCAUSAL SAMPLE-LEVEL ANALYSIS

F APPENDIX F: RUNTIME AND RESOURCE DETAILS

EconCausal evaluation required 74 minutes for the baseline and 81 minutes for the finetuned model (BF16, batch size 4, RTX 5090). Corr2Cause evaluation required approximately 3 minutes per run. MMLU evaluation required approximately 20 minutes per run.

Table 6: Corr2Cause accuracy by causal template type. The finetuned model shows the largest gains on structurally complex templates.

Template	GT True%	Baseline True%	Baseline Acc	Finetuned Acc	Δ
child	0.0%	19.1%	80.9%	100.0%	+19.1pp
non-child descendant	5.7%	92.7%	13.0%	94.3%	+81.3pp
non-parent ancestor	11.3%	96.9%	14.4%	87.2%	+72.8pp
has_confounder	8.8%	97.4%	11.9%	54.9%	+43.0pp
parent	33.5%	43.8%	62.9%	66.5%	+3.6pp
has_collider	33.1%	98.4%	34.7%	44.6%	+9.8pp

Table 7: EconCausal: top regression patterns by ground truth $\rightarrow$ baseline prediction $\rightarrow$ finetuned prediction.

Pattern	T1 Econ	T1 Fin	T2	T3	Description
$+ \rightarrow$ mixed	96	95	9	34	Hedging positive effects
$+ \rightarrow -$	37	41	5	36	Flipping positive to negative
$+ \rightarrow$ none	25	20	–	–	Nullifying positive effects
$- \rightarrow$ mixed	8	–	–	11	Hedging negative effects
$- \rightarrow +$	6	–	–	–	Flipping negative to positive